

# Chat Filter Accuracy Audit

The filter matches blocked terms as **unanchored substrings**, and the term list is **predominantly Brazilian Portuguese**. Two-letter Portuguese entries applied across all eight locales censor ordinary vocabulary in every language the game ships in.

|                    |                           |                              |                              |
|--------------------|---------------------------|------------------------------|------------------------------|
| thank<br>DELIVERED | than <b>ks</b><br>REFUSED | do <b>cu</b> ment<br>REFUSED | se <b>cu</b> rity<br>REFUSED |
|--------------------|---------------------------|------------------------------|------------------------------|

cu is Portuguese for anus; ks is a literal two-character entry. Neither has a word-boundary condition, so both fire inside any word that happens to contain them.

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**15,307**

MESSAGES SENT

**343**WORDS CONFIRMED  
CENSORED**8,414**

ENGLISH WORDS AFFECTED

**16**

RULES IDENTIFIED

**Content notice.** Establishing what the filter holds requires naming the terms it matches, so this report quotes profanity and slurs. They appear as evidence, not as commentary.

Black-box measurement against the live service, 11–12 September 2026. No client modification, no packet injection, no server or source access — the audit observes exactly what a player observes. All 15,307 probe records are published with this report.

## Blocked terms have no word boundary

A term fires wherever its letters appear. This was isolated by probing the sequence inside inert padding, so a refusal can only be attributed to the sequence itself rather than to the word that revealed it.

| PROBE  | RESULT    | PROBE  | RESULT    | ESTABLISHES                        |
|--------|-----------|--------|-----------|------------------------------------|
| cu     | refused   | ks     | refused   | the sequence is on the list        |
| xxcuxx | refused   | xxksxx | refused   | it fires inside an unrelated word  |
| xxcXX  | delivered | xxkxx  | delivered | the first letter alone does not    |
| xxUxx  | delivered | xxSxx  | delivered | the second letter alone does not   |
| xxucxx | delivered | xxskxx | delivered | order matters — a literal sequence |

Across the **10,627 words the filter accepted**, not one contains **cu** or **ks**. There are no exceptions in either direction.

## Shorter entries do exponentially more damage

Ordinary English words affected, by rule. Because matching is unanchored, an entry's blast radius grows as its length shrinks.

|       |             |       |                             |
|-------|-------------|-------|-----------------------------|
| cu    | <div></div> | 5,016 | document, discuss, security |
| nb    | <div></div> | 956   | number, inbox, unbind       |
| meter | <div></div> | 695   | parameter, diameter         |
| anta  | <div></div> | 419   | advantage, santa, fantastic |
| ks    | <div></div> | 408   | thanks, tasks, books, works |
| pus   | <div></div> | 230   | campus, octopus, push       |
| pica  | <div></div> | 227   | typical, tropical           |
| puta  | <div></div> | 95    | reputation, computation     |

Eight further rules account for the remainder.  two-character entry  longer entry

## Coverage is uneven by language, not uniformly strict

Twenty-seven Portuguese profanity terms were tested and **all twenty-seven were refused**, down to low-frequency items like *arrombado* and *otario*. Tested identically:

|           |           |           |           |
|-----------|-----------|-----------|-----------|
| bitch     | rape      | slut      | whore     |
| DELIVERED | DELIVERED | DELIVERED | DELIVERED |

Both halves share one cause: a term list that has not been reviewed against the locales it is applied to.

## Every market, and the game's own interface text

The refusal rate was measured separately against the common vocabulary of each shipped language. It is uniform — this is not an English-only problem.

|            |                        |      |                            |
|------------|------------------------|------|----------------------------|
| Indonesian | <div><div></div></div> | 4.8% | aksi, manusia, perubahan   |
| Thai       | <div><div></div></div> | 3.0% | romanised interface tokens |
| English    | <div><div></div></div> | 2.8% | thanks, document, security |
| Portuguese | <div><div></div></div> | 2.8% | acumulado, vermelho        |
| Vietnamese | <div><div></div></div> | 2.2% | anh, thanh, linh, kho      |

**The service censors words from its own shipped localisation files.** A player who types a word they can read on screen is told their speech is inappropriate.

## What the player experiences

The message is not delivered and the sender sees a modal stating the speech is inappropriate. Nothing indicates which word caused it, and the offending sequence is not visible as a word — so the rule cannot be learned and cannot be avoided. Repeated refusals of ordinary vocabulary read as the game being broken rather than as moderation.

## Why it carries a support cost

| PROPERTY                          | CONSEQUENCE                                                                       |
|-----------------------------------|-----------------------------------------------------------------------------------|
| Affects high-frequency vocabulary | thanks is among the most common words in social game chat, and it is refused      |
| Silent and unattributable         | Players cannot infer the rule, so behaviour does not adapt                        |
| Not individually actionable       | Each report names a different ordinary word produced by the same handful of rules |
| Asymmetric by market              | Indonesian shows the highest rate of the five vocabularies measured               |

## Confidence in these figures

| FIGURE                              | BASIS                                                                                                      |
|-------------------------------------|------------------------------------------------------------------------------------------------------------|
| <b>343</b> words confirmed censored | Measured. Each required two independent refused observations against a measured 1.2% per-probe error rate. |
| <b>8,414</b> English words affected | Derived. Confirmed rules projected across a dictionary; these specific words were not each sent.           |
| <b>16</b> rules                     | A lower bound. Entries whose letters did not occur in the 11,868 strings tested would not have surfaced.   |

## Four options. None of them blocks less.

Every option below removes false positives only — no entry is taken off the block list. They are independent and can ship in any order.

|          | APPROACH                     | TOUCHES           | RESIDUAL RISK                          | EFFORT    |
|----------|------------------------------|-------------------|----------------------------------------|-----------|
| <b>A</b> | Ship the confirmed allowlist | Data only         | Words outside the tested set           | Hours     |
| <b>B</b> | Add the rescue shim          | One call site     | Names and slang not in a dictionary    | 1 day     |
| <b>C</b> | Minimum-length boundary rule | Matcher condition | Long entries still match as substrings | 1–2 days  |
| <b>D</b> | Replace the matcher          | Component swap    | None of this class                     | 1–2 weeks |

### A · Allowlist — no code change

If the filter already consults an exception list, this is a data deployment. `words-to-allow.txt` holds the 343 confirmed words, one per line, each annotated with the rule responsible.

### C · Boundary rule — best value per line

One condition addresses the cause rather than the symptoms. Long entries such as `caralho` behave exactly as they do today, so strength against substantial terms is untouched.

**Entries shorter than five characters match only as whole words.** Entries of five characters or more are unchanged.

Under this rule `cu` still refuses `cu` and stops refusing `document`; `ks` still refuses `ks` and stops refusing `thanks`.

## Recommended rollout

- 1 Shadow.** Evaluate the new decision alongside the current one and log where they differ, without changing what players see.
- 2 Check the direction of every difference.** All differences must be messages previously refused that would now be delivered. A message newly *refused* indicates a wiring error, not a tuning question — none of these options adds blocking.
- 3 Enable.** A and B may ship independently and in either order.

## Three lines at one call site

Call the shim where the existing filter has decided to block and knows the matched span. If that span sits inside ordinary vocabulary, suppress the block. Your matching logic, word list, severity rules and protocol are all unchanged.

```
# once, at startup
from rescue import Rescue
rescue = Rescue("allowlist-en.txt")

# inside your filter, where you were about to block:
if rescue.is_rescued(text, match_start, match_end):
    pass          # ordinary word – deliver the message
else:
    block()       # unchanged behaviour
```

**Cost:** 473,532 entries as a hash set, loaded once. Lookup is a single hash probe — not measurable against a network round trip.  
**Why it works without knowing your list:** you assert what is legitimate rather than identifying which entries are at fault, so it holds however the list changes.

## The boundary rule, language-agnostic

```
// term matched at [start, end) in the message
func accept(term, message, start, end) {
    if (len(term) >= MIN_SUBSTRING_LEN) return true;    // 5 – unchanged

    // short term: require a word boundary on both sides
    before = (start == 0)          ? null : message[start-1];
    after  = (end == len(message)) ? null : message[end];
    return !isLetter(before) && !isLetter(after);
}
```

If your matcher is a compiled automaton, tag short patterns at build time and apply the boundary test only to those, so throughput is unaffected for the rest.

## What ships in this package

| PATH                        | FORMAT                     | SIZE    | USE                      |
|-----------------------------|----------------------------|---------|--------------------------|
| findings/words-to-allow.txt | one word per line          | 343     | Option A payload         |
| findings/affected-words.txt | one word per line, by rule | 8,414   | Option A, broader set    |
| fix/allowlist-en.txt        | one word per line          | 473,532 | Option B corpus          |
| fix/rescue.py               | Python 3, no dependencies  | 138 In  | Option B shim            |
| engine/                     | Python 3, no dependencies  | ~650 In | Option D matcher         |
| vectors/golden.jsonl        | JSONL                      | 29      | Conformance for any port |

## Confirm the defect in ten seconds

Send thank in game, then thanks. The first is delivered, the second refused. This requires nothing from this package.

### Must become deliverable

| INPUT      | REFUSED BY |
|------------|------------|
| thanks     | ks         |
| document   | cu         |
| number     | nb         |
| advantage  | anta       |
| parameter  | meter      |
| reputation | puta       |

### Must stay refused

| INPUT     | NOTE                            |
|-----------|---------------------------------|
| cu        | the entry itself, as a word     |
| ks        | the entry itself, as a word     |
| puta      | short entry, still matches      |
| caralho   | long entry, rule does not apply |
| arrombado | long entry, unaffected          |

If any of these becomes deliverable the change was wired incorrectly — no option removes an entry from the block list.

```
python3 tests/test_all.py      # 53 tests
python3 tools/conformance.py  # 29 language-agnostic vectors
python3 tools/check_numbers.py # every figure here traces to the evidence
```

## 06 · METHOD & LIMITATIONS

Messages were sent through the retail client into a private party channel, one at a time, read back from chat history. Three controls govern what reached this report:

| CONTROL                                                     | WHY                                                                                                   |
|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| A substring is not reported until isolated in inert padding | That cucumber is refused does not identify the responsible part                                       |
| Two independent refused observations required               | Measured per-probe error rate is 1.2% (6 spurious in 491), so one reading is not a result             |
| An unsent probe counts as untested, not as passing          | All but 15 were retried to a real verdict; a false-negative audit over 250 words returned zero misses |

| LIMITATION                      | CONSEQUENCE                                                           |
|---------------------------------|-----------------------------------------------------------------------|
| Rule set is a lower bound       | There may be further entries the tested vocabulary did not reveal     |
| Dictionary counts are derived   | 8,414 is a projection; the 343 measured words are kept separately     |
| Point-in-time                   | Describes the service on 11–12 September 2026; a list edit changes it |
| Non-Latin locales under-covered | Korean, Chinese and Japanese were not swept; no conclusion is offered |

Every quantity here derives from `findings/findings.json` and is checked by `tools/check_numbers.py`, which fails if a published figure is not supported by the evidence. All 15,307 probe records: `findings/raw-logs/`.